

The Status of the Proca Sector in the WIN Paradigm: Reconciling ENTROPIX and ENTROPIA-EFT

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

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Abstract

Two WIN Paradigm papers, *ENTROPIX* [32] and *ENTROPIA-EFT* [31], make conflicting statements about the status of the Proca sector of the vacuum effective field theory. ENTROPIX lists the Proca Lagrangian A_1 and the steady-state flux A_2 as closed derivations from a substrate $U(1)$ phase symmetry. ENTROPIA-EFT lists A_1 and A_2 as effective ansätze awaiting a first principles derivation. We resolve the conflict by distinguishing three separate claims: the global $U(1)$ phase symmetry of the 14-mode multiplet (which is derived); the conserved Noether current J_S^μ and its conservation law $\nabla_\mu J_S^\mu = 0$, which is A_2 (derived given the $U(1)$ symmetry); and the Proca Lagrangian A_1 itself, which requires promoting the global $U(1)$ to a local gauge symmetry and assuming a specific mass term, neither of which follows from the $U(1)$ alone. We prove that the global $U(1)$ symmetry constrains the current but does not force the Proca form of its dynamics. The non minimal coupling $\xi R s_\mu s^\mu$ is derived from the trace anomaly given the Proca form. We conclude that the correct status of the Proca sector is: $U(1)$ derived, A_2 derived, A_1 partially derived (kinetic term standard; non-minimal coupling derived; mass term ansatz). We catalogue the specific corrections to ENTROPIX and confirm that ENTROPIA-EFT's statements are already correct. This paper closes Tier 0 of the WIN open problem list.

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1 Introduction

The WIN Paradigm describes the vacuum as an effective field theory built on the 8×8 torus substrate [10, 11, 12]. Two papers in the series discuss the effective-field-theory sector:

- *ENTROPIX: A Master Entropy Selection Framework for Vacuum Structure and Gravitational Coupling* [32] states, in §7.3, that the Proca Lagrangian A_1 is a closed derivation from the substrate $U(1)$ phase symmetry, and lists the Proca mass m_s as derived.
- *ENTROPIA-EFT: The Effective Field Theory of the DME Vacuum* [31] states, in §8.4 and Table 3, that the Proca Lagrangian A_1 and the steady-state flux A_2 are effective ansätze, and that the $U(1)$ origin is under investigation.

These statements directly contradict. The purpose of this paper is to resolve the conflict and to state precisely what is derived and what is not.

The resolution turns on a distinction that neither paper states explicitly:

A global $U(1)$ symmetry of a field theory does not imply a specific dynamics for its Noether current.

The substrate $U(1)$ symmetry is a symmetry of the 14-mode multiplet under a global phase rotation. It gives a conserved Noether current. It does *not* imply that this current is governed by a Proca action. The Proca form is a natural ansatz, consistent with the $U(1)$ symmetry, but not forced by it.

The distinction is familiar from effective field theory [6, 5, 7]: a global symmetry fixes the current, its conservation law, and the corresponding Ward identities, but it does not fix the operator content of the effective action. The specific dynamics of the current must be either measured or derived from a more fundamental theory. The Proca sector of the WIN Paradigm is a clean illustration of this general lesson, and the reconciliation reported here applies to any framework in which a global $U(1)$ is proposed as the origin of an emergent vector field.

We prove the distinction rigorously, then state the corrected status of A_1 , A_2 , and the $U(1)$ symmetry. We close by cataloguing the specific corrections to ENTROPIX. ENTROPIA-EFT's statements are already correct and require no change.

1.1 Organization

§2 states the two conflicting claims. §3 resolves the conflict by separating three claims that both papers conflate. §4 states the corrected status of the Proca sector. §5 discusses what would close the derivation of A_1 . §6 catalogues the corrections to prior papers. §8 concludes.

2 The two conflicting statements

2.1 ENTROPIX

ENTROPIX [32] §7.3 lists five “closed derivations.” Item 2 is:

Derivation of A_1 (Proca) from the substrate $U(1)$ phase symmetry. Proca mass:

$$m_s = \frac{M_P}{N} \exp\left(-\frac{d+1}{d-1}N\right) = 9.0247 \dots \times 10^{-21} \text{ eV}.$$

In the predictions table (ENTROPIX Table 1) the Proca mass is listed as a derived prediction of the framework. ENTROPIX §10.1 lists the Proca mass window as a falsification channel, on the assumption that its value is derived.

2.2 ENTROPIA-EFT

ENTROPIA-EFT [31] §8.4 is explicit:

The Proca Lagrangian (44) and the steady-state flux condition are currently treated as effective ansätze, not as derived results. The reason for this is that no unique derivation of the $U(1)$ symmetry from the substrate has yet been established.

In the assumption audit (ENTROPIA-EFT Table 3), the Proca Lagrangian A_1 , steady-state flux A_2 , and backreaction tensor $\Delta_{\mu\nu}$ are all listed as *Assumed*. Only the Einstein equations and the Fisher metric are listed as derived from the substrate.

2.3 The contradiction

The two papers cannot both be right. Either A_1 is a closed derivation or it is an effective ansatz. Either the Proca mass is a prediction of the framework or it is a conditional consequence of an ansatz.

3 Resolution: what the $U(1)$ symmetry does and does not imply

We resolve the conflict by separating three distinct claims that both papers conflate.

3.1 The substrate $U(1)$ phase symmetry

Theorem 1 ($U(1)$ phase symmetry is derived). *The 14-mode multiplet of the 8×8 torus carries an exact global $U(1)$ phase symmetry generated by the phase rotation*

$$v_k(x) \mapsto e^{i\theta} v_k(x) \tag{1}$$

for every mode $k \in \mathcal{K}_{14}$ and every $\theta \in \mathbb{R}$.

Proof. The 14 modes of the multiplet are carried by the complex plane-wave basis $v_k(x) = e^{i\pi k \cdot x/4}$ for $k \in \mathcal{K}_{14}$, where $\mathcal{K}_{14}$ is the mode set at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian [11, 12]. The Laplacian acts on each plane wave as a scalar $\lambda(k)$, and the phase rotation $v_k \mapsto e^{i\theta} v_k$ commutes with the Laplacian and preserves the multiplet. Hence it is a global symmetry of the free substrate theory. $\square$

The $U(1)$ symmetry is a genuine derived feature. It is candidate (iii) of ENTROPIA-EFT §8.3: the phase symmetry of the mode wavefunctions.

3.2 The Noether current and A_2

Proposition 2 (Noether current and A_2 are derived). *The $U(1)$ symmetry of Theorem 1 implies a conserved Noether current*

$$J_S^\mu = i \sum_{k \in \mathcal{K}_{14}} (v_k^* \partial^\mu v_k - v_k \partial^\mu v_k^*), \quad (2)$$

satisfying $\partial_\mu J_S^\mu = 0$ in flat spacetime. In a curved background,

$$\nabla_\mu J_S^\mu = 0. \quad (3)$$

Equation (3) is exactly the condition A_2 of ENTROPIA-EFT.

Proof. Standard Noether construction [8, 9] applied to the global $U(1)$ phase symmetry of Theorem 1. In curved spacetime, the flat space conservation law $\partial_\mu J^\mu = 0$ becomes the covariant divergence condition $\nabla_\mu J^\mu = 0$ upon minimal coupling. $\square$

So the $U(1)$ symmetry is derived, and A_2 is a direct consequence of it. Both statements are correct.

3.3 The Proca Lagrangian A_1 is not implied by $U(1)$

The conflict is entirely about whether the Proca Lagrangian

$$\mathcal{L}_S = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_s^2 s_\mu s^\mu + \xi R s_\mu s^\mu \quad (4)$$

is derived from the substrate.

Theorem 3 (The Proca Lagrangian is not implied by $U(1)$). *The Proca Lagrangian (4) is not a consequence of the global $U(1)$ phase symmetry of Theorem 1. Establishing (4) requires two additional inputs:*

1. **Gauging.** *Promoting the global $U(1)$ phase rotation $v_k \mapsto e^{i\theta} v_k$ to a local rotation $v_k \mapsto e^{i\theta(x)} v_k$, which requires the introduction of a gauge connection A_μ , and the identification of A_μ with the physical vector field s_μ .*
2. **Mass term.** *Assuming the vector field s_μ has a mass m_s and that its dynamics is governed by the Proca action, rather than by a massless (Maxwell) action or a higher-derivative alternative.*

Neither input is a consequence of the global $U(1)$ symmetry alone. Both are additional assumptions.

Proof. A global symmetry constrains the current J_S^μ but does not specify its dynamics. The current is a composite operator built from the fields v_k ; its dynamics is determined by the underlying theory of the v_k , not by the symmetry alone. To promote the current to a fundamental field s_μ with its own dynamics requires the gauging step (1), which introduces a new degree of freedom not present in the substrate $U(1)$ symmetry.

Given the gauging step, the kinetic term $-\frac{1}{4}F^2$ is standard, but the mass term $\frac{1}{2}m_s^2 s^2$ requires the additional assumption (2) that the vector field is massive. A massless Abelian gauge field has no longitudinal polarization, and the massless action $-\frac{1}{4}F^2$ is invariant under the local gauge transformation $s_\mu \mapsto s_\mu + \partial_\mu \lambda$. A mass term $\frac{1}{2}m_s^2 s_\mu s^\mu$ explicitly breaks this gauge invariance and is prohibited in the strict gauge theory unless it is generated by a mechanism that preserves the gauge structure.

The standard mechanism for gauge invariant mass generation is the Stückelberg construction [4, 1]: introducing an additional scalar field ϕ that transforms as $\phi \mapsto \phi - m_s \lambda$ under the local $U(1)$, so that the combination $s_\mu + \frac{1}{m_s} \partial_\mu \phi$ is gauge-invariant. The Proca mass term then arises from the Stückelberg Lagrangian by gauge fixing $\phi = 0$, but this requires the explicit introduction of the Stückelberg field, which is not a consequence of the substrate $U(1)$ symmetry.

Alternatively, the mass term can arise from the Higgs mechanism [2, 3], in which case a scalar field acquires a vacuum expectation value and generates m_s dynamically. Neither the Stückelberg field nor the Higgs field is present in the substrate $U(1)$ construction; introducing either is an additional assumption beyond the phase symmetry of Theorem 1.

The non-minimal coupling $\xi R s^2$ is not independent: given the gauging step (1) and the mass assumption (2), it is required for the trace anomaly to contribute to the trace of the stress tensor, as shown in ENTROPIA-EFT §7.1. But its derivation presupposes (1) and (2). In the language of EFT operator classifications [6, 5, 7], the Proca Lagrangian corresponds to a specific Wilson coefficient pattern that is not selected by the global symmetry alone. $\square$

Remark 4. The Proca form (4) is a natural ansatz. It is the unique renormalizable action for a massive vector field with a canonical kinetic term, and it is what the Stückelberg or Higgs mechanism produces in the appropriate limits. It is not, however, forced by the substrate $U(1)$ symmetry. Deriving it from first principles requires one of the mass-generation mechanisms discussed in the proof above.

3.4 Summary of the resolution

The resolution is now clean:

Statement	Derived?	Source
Global $U(1)$ phase symmetry	Yes	Theorem 1
Noether current J_S^μ	Yes	Proposition 2
Conservation law A_2	Yes	Proposition 2
Kinetic term $-\frac{1}{4}F^2$	Standard	Given gauging
Non-minimal coupling $\xi R s^2$	Conditional	Trace anomaly
Mass term $\frac{1}{2}m_s^2 s^2$	No	Ansatz
Proca Lagrangian A_1	Partially	Theorem 3

Table 1: Status of each element of the Proca sector.

4 Corrected status of the Proca sector

We now state the correct status of the Proca sector.

1. **The global $U(1)$ phase symmetry is derived** from the plane-wave basis of the 14-mode multiplet (Theorem 1).
2. **The Noether current J_S^μ is derived** from the $U(1)$ symmetry (Proposition 2).
3. **The steady-state flux condition A_2 is derived** from current conservation (Proposition 2, Eq. (3)).

4. **The Proca Lagrangian A_1 is partially derived.** The kinetic term $-\frac{1}{4}F^2$ is standard given gauging. The non-minimal coupling $\xi R s^2$ is required by the trace anomaly given the Proca form. The mass term $\frac{1}{2}m_s^2 s^2$ is an ansatz that is not implied by the $U(1)$ symmetry (Theorem 3).
5. **The Proca mass m_s is a conditional prediction.** It is derived from the EDF running given the Proca ansatz and the EDF fixed point normalization $\gamma_f = 1$. It is not a prediction of the substrate alone.

5 What would close the derivation of A_1

Two paths would promote A_1 from a partial derivation to a full one:

1. **Derive the gauging.** Show that the substrate $U(1)$ symmetry is necessarily local, i.e., that the effective theory of the 14-mode multiplet at low energy requires promoting the global symmetry to a local one. This could arise, for example, if the substrate contains an emergent gauge field, or if the phase symmetry is part of a larger gauge structure.
2. **Derive the mass term.** Show that the composite operator J_S^μ has a nontrivial two-point function with a mass gap at low energy, or that the underlying 14-mode theory generates a mass term for the current. This could arise from the DME modulus ϵ coupling to the current, if that coupling generates a Proca mass dynamically, either via a Stückelberg mechanism [4] or via a Higgs-like condensation.

Until one of these is established, A_1 should be regarded as an effective framework, consistent with the $U(1)$ symmetry but not forced by it.

6 Corrections to prior papers

6.1 Corrections to ENTROPIX

The following corrections should be applied to ENTROPIX [32]:

Current statement	Corrected statement
§7.3 item 2: “Derivation of A_1 (Proca) from the substrate $U(1)$ phase symmetry”	“Derivation of the $U(1)$ phase symmetry and the conserved Noether current J_S^μ from the substrate. Identification of the Proca Lagrangian A_1 as a natural ansatz consistent with the $U(1)$ symmetry, with the mass term treated as an additional input.”
§10.1 predictions table: Proca mass m_s listed as derived prediction	Proca mass listed as a conditional prediction of the EDF running given the Proca ansatz.
Table 3 (one-line summary): “ENTROPIX/MESA: master selection framework, closed sub-sector derivations”	“ENTROPIX/MESA: master selection framework. Closed: w , Einstein equations, H_0 boost, γ regulator independence, $U(1)$ phase symmetry, A_2 . Conditional: A_1 (Proca Lagrangian), Proca mass.”

Table 2: Corrections to ENTROPIX.

6.2 Corrections to ENTROPIA-EFT

No corrections. ENTROPIA-EFT [31] already states the correct position in §8.4 and Table 3.

6.3 Cross-reference additions

Both papers should be updated to cross-reference this reconciliation:

- ENTROPIX: add a footnote in §7.3 referring to this paper for the partial-derivation status of A_1 .
- ENTROPIA-EFT: add a reference in §8.4 citing this paper as the formal statement of the position.

7 Discussion

7.1 The general lesson

The conflict between ENTROPIX and ENTROPIA-EFT is not unique to the WIN Paradigm. It reflects a general ambiguity that arises whenever a global symmetry is proposed as the origin of an emergent vector field:

A global symmetry constrains the current. It does not constrain the current's dynamics. The dynamics must be derived from a more fundamental theory, or assumed as an ansatz.

In the case of the substrate $U(1)$ phase symmetry of the 14-mode multiplet, the following are all consistent with the symmetry:

- a Maxwell action $-\frac{1}{4}F^2$ for a massless gauge field;
- a Proca action $-\frac{1}{4}F^2 + \frac{1}{2}m_s^2 s^2$ for a massive vector field;
- a Stückelberg action $-\frac{1}{4}F^2 + \frac{1}{2}m_s^2(s_\mu + \frac{1}{m_s}\partial_\mu\phi)^2$ with an additional scalar;
- a Higgs action $-\frac{1}{4}F^2 + |D_\mu H|^2 - V(H)$ with a scalar doublet;
- a higher-derivative action $-\frac{1}{4}F^2 + \frac{c}{M^2}(\partial_\mu F^{\mu\nu})^2 + \dots$ with new Wilson coefficients.

The Proca form is the simplest of these for a massive vector field. It is not selected by the $U(1)$ symmetry alone.

7.2 Positioning

As an internal white paper in the Entropia Research Institute series, this paper performs a vital self-correction: it identifies an inconsistency between two prior papers and resolves it in favor of the more careful position. This is the kind of editorial work that keeps the framework auditable and prevents drift.

For external publication, the paper is best framed as a short methodological note on the constraints imposed by global symmetries on emergent vector fields, using the WIN Proca sector as a worked example. The general lesson — that a global symmetry fixes the current but not its dynamics — applies to any framework in which emergent gauge fields are proposed from discrete or combinatorial substrates.

8 Conclusion

We have resolved the conflict between ENTROPIX and ENTROPIA-EFT on the status of the Proca sector. The resolution distinguishes three separate claims that both papers conflate:

1. The global $U(1)$ phase symmetry of the 14-mode multiplet is derived.
2. The conserved Noether current and the steady-state flux condition A_2 are derived given the $U(1)$ symmetry.
3. The Proca Lagrangian A_1 is not implied by the $U(1)$ symmetry; it requires the additional inputs of gauging and mass term.

We have corrected ENTROPIX’s §7.3, §10.1, and Table 3 to reflect this. ENTROPIA-EFT’s §8.4 and Table 3 already state the correct position and require no change.

The corrected status of the Proca sector is:

- $U(1)$ derived.
- A_2 derived.
- A_1 partially derived: kinetic term standard; non-minimal coupling derived given the Proca form; mass term ansatz.
- Proca mass m_s conditional on the Proca ansatz and the EDF fixed-point normalization.

This closes Tier 0 of the WIN open problem list. The framework’s internal consistency is restored by adopting the more careful position of ENTROPIA-EFT as the correct statement.

Two paths would close the derivation of A_1 fully: deriving the gauging of the substrate $U(1)$, or deriving the mass term dynamically via a Stückelberg or Higgs-like mechanism. Both remain open.

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